Audio course on steroids during pregnancy. Episode nine. Fetal and neonatal effects, programming, and breastfeeding.

Companion reading handout for simultaneous listening.
This episode is about balance.

A single well-timed fetal steroid course before early preterm birth is one of the most effective interventions in obstetrics. At the same time, glucocorticoids are developmental signals, not inert drugs. Unnecessary, repeated, prolonged, or wrong-target exposure deserves caution.

The mature position is not steroid fear and not steroid casualness. It is precision.

Start with the proven short-term benefit.

For a fetus likely to be born early preterm, antenatal betamethasone or dexamethasone reduces respiratory distress syndrome, intraventricular hemorrhage, necrotizing enterocolitis, and neonatal mortality. That is why we give the course when timing and indication are right.

Now the short-term tradeoffs.

Steroids can worsen maternal hyperglycemia. In the late preterm setting, betamethasone increases neonatal hypoglycemia. Steroids can also transiently affect fetal behavior, reducing fetal breathing movements and fetal heart rate reactivity. A resident interpreting an NST or BPP soon after steroids should remember that the test may look temporarily less reassuring because of the medication.

That does not mean ignore fetal testing. It means do not interpret testing without context.

Now first-trimester structural risk.

Older literature raised concern about systemic corticosteroids and oral clefts. UKTIS is helpful because it frames the evidence with balance. Some animal data and some human studies suggest an association during organogenesis, but most human data do not show a clear increase. If there is an increase, the absolute risk appears small.

So the counseling line should be neither dismissive nor frightening.

Do not say, there is no concern at all.

Do not say, steroids are a major known teratogen.

Say that some studies raised a possible small signal for oral clefts, but most human data are reassuring, and serious maternal disease may be more dangerous than appropriately selected treatment. Use the lowest effective dose and prefer lower fetal-transfer steroids for maternal disease when clinically suitable.

Now fetal growth.

Repeated or chronic glucocorticoid exposure raises more concern for fetal growth restriction than a single well-timed course. But growth restriction is complicated. The drug may contribute. The disease may contribute. The placenta may contribute. Lupus, severe asthma, active IBD, preeclampsia, placental insufficiency, malnutrition, and inflammation can all affect fetal growth.

The resident should not automatically blame the steroid or automatically dismiss the steroid. Ask, is growth risk from the drug, the disease, the placenta, or all three?

Now neonatal adrenal suppression.

High-dose or prolonged maternal systemic steroid exposure near delivery can suppress neonatal adrenal function, depending on the steroid, dose, duration, placental transfer, and timing. The risk is generally remote with usual prednisone or methylprednisolone maternal therapy, because placental metabolism limits fetal exposure, but it is not impossible in high-dose, prolonged, or fluorinated exposure settings.

Neonatology should know about prolonged or high-dose maternal steroid use, especially if exposure occurred near delivery or involved a steroid that crosses the placenta efficiently.

Now long-term programming.

Gabbe's developmental origins chapter is central here. Glucocorticoids influence fetal organ maturation, but they also influence developmental programming. Fetal tissues express glucocorticoid receptors from mid-gestation onward. Glucocorticoids can affect gene expression through epigenetic mechanisms, H P A axis development, brain development, metabolic function, cardiovascular biology, renal development, and immune development.

Human follow-up after antenatal betamethasone has generally been reassuring in many domains, but signals have been reported for insulin response and stress-axis physiology. Repeated courses have raised stronger concern, including reduced growth measures and possible behavioral or neurodevelopmental associations in some studies.

Here is the teaching frame.

A single indicated course has strong neonatal benefit. The uncertainty belongs mainly to unnecessary exposure, repeated courses, late preterm expansion, and fetuses exposed but ultimately born at term.

This is why precision matters so much.

Now breastfeeding.

Prednisone and prednisolone are generally compatible with breastfeeding. Inhaled steroids are compatible. Hydrocortisone replacement is compatible. Gabbe's drug chapter reports very low prednisone transfer into breast milk, even at eighty milligrams per day, far below the infant's endogenous cortisol exposure.

Still, for higher prednisone or prednisolone doses, some clinicians time breastfeeding to reduce infant exposure. Gabbe's connective tissue chapter suggests that when prednisone doses exceed twenty milligrams per day, delaying feeding for about four hours after the dose may be prudent.

Premature or medically fragile neonates may justify more individualized lactation pharmacology review.

Now integrate the risk language.

For threatened preterm birth at thirty weeks, the fetal benefit is large and immediate. The appropriate counseling emphasizes lung maturation, prematurity risk reduction, maternal glucose monitoring, and neonatal glucose monitoring.

For a lupus flare at ten weeks, the counseling is different. The steroid is maternal therapy. Use a prednisolone-type drug when appropriate. Discuss the possible small oral-cleft signal without exaggeration. Explain that uncontrolled lupus can harm both mother and fetus.

For repeated courses, the counseling changes again. A rescue course may be considered before thirty four weeks when the prior course was more than fourteen days earlier and preterm birth risk is again real. But repeated exposure is not casual. It requires a renewed risk-benefit decision.

The resident reflex is this: the concern is not a single indicated course. The concern is unnecessary, repeated, prolonged, or wrong-target exposure.

Use steroids when the target and timing are right.

Monitor the predictable tradeoffs.

Counsel honestly.

And remember that fetal risk is rarely only the drug. It is the drug, the disease, the placenta, the timing, the dose, and the reason treatment was needed in the first place.